

LinFill-120 Vertical Syringe Filling Station

LinFill Series · Elara Automation GmbH
MQTT Interface Specification

This specification defines the complete MQTT 3.1.1 communication interface of the LinFill-120 Vertical Syringe Filling Station. It is intended for control engineers, SCADA/MES integrators, and Unified Namespace architects who need to connect the station to a supervisory system or data infrastructure.

All information in this document refers to firmware version 1.5.0 and later, which introduced the correlated command/response topic pattern described in Sections 3-5. Firmware versions prior to 1.5.0 used a single shared command topic and are not compatible with this specification -- see EA-LC-LF120-EN Lifecycle Information Document for the firmware upgrade path.

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About This Document

This document has been prepared by Elara Automation GmbH and is intended for qualified technical personnel responsible for the installation, commissioning, operation, and maintenance of the equipment described herein. Readers are expected to be familiar with general principles of industrial automation, electrical safety, and network-connected laboratory equipment.

The document is structured as a reference manual. Sections are broadly ordered to follow the typical lifecycle of the equipment — from first delivery through to end-of-life disposal — but individual sections may be read in isolation. Cross-references are provided where information in one section depends on material covered elsewhere.

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Conventions Used in This Document

The following conventions are used throughout:

Notation	Meaning
{serialNumber}	The unit's serial number as printed on the rear label (e.g. EA-PS-2024-00391)
{base}	The site-specific topic prefix, e.g. NN/Nybrovej/InnoLab/Filling, configured during commissioning
Uuid	A client-generated correlation identifier included in every command payload and echoed back in the matching response payload
NOTE	Supplementary information that aids understanding but is not safety-critical
CAUTION	Risk of equipment damage or process disruption if instruction is not followed
WARNING	Risk of personal injury if instruction is not followed

Related Documents

Document No.	Title
EA-LC-LF120-EN	LinFill-120 Lifecycle Information Document
EA-LN-SPL01-EN	SPL-01 Line Description & Bill of Materials

1 Introduction

MQTT (Message Queuing Telemetry Transport, ISO/IEC 20922:2016) is a lightweight, publish-subscribe messaging protocol well suited to embedded devices with constrained resources and unreliable network connections. The LinFill-120 acts as an MQTT client and connects to a broker provided by the facility. It does not include an embedded broker; a separate broker (e.g. HiveMQ, Eclipse Mosquitto, EMQX) must be available on the facility network.

Every command the station accepts is issued and acknowledged as a correlated request/response pair: a client publishes a JSON command to a dedicated `{base}/CMD/<Action>` topic carrying a unique "Uuid" value, and the station publishes a JSON result to the corresponding `{base}/DATA/<Action>` topic once the command has been processed, echoing the same "Uuid" so the client can match the response to its request. This request/response pattern is used consistently across all Elara Automation InnoLab-integrated equipment.

The topic prefix — referred to as `{base}` throughout this document — is site-specific and configured during commissioning. There is no default value; the prefix must be agreed between the equipment owner and the system integrator before commissioning and recorded in the site namespace register (e.g. `NN/Nybrovej/InnoLab/Filling` for a station installed in the Nybrovej InnoLab).

2 Transport and Connection Parameters

The MQTT connection is carried over TCP using the device's 2.4 GHz WiFi interface (IEEE 802.11 b/g/n). The broker port number depends on whether TLS is in use. For development or isolated lab networks, plain TCP on port 1883 is acceptable; for production environments Elara Automation strongly recommends MQTTS on port 8883 with mutual TLS certificate authentication.

Protocol version	MQTT 3.1.1 (ISO/IEC 20922:2016)
Transport	TCP/IP over IEEE 802.11 b/g/n WiFi, 2.4 GHz band
Default port (plain)	1883
Recommended port (TLS)	8883 with TLS 1.2 or higher
Client ID	linfill120-{serialNumber} — e.g. linfill120-EA-LF-2024-00472
Clean Session	false (persistent session; QoS 2 subscriptions survive reconnects)
QoS (all topics)	2 (exactly once) — required for command/response correlation, see Section 4
Keep-alive interval	60 seconds (broker will detect offline device after 90 s)
Reconnect strategy	Exponential back-off starting at 1 s, doubling up to 60 s maximum
Authentication	Username / password pair, set during provisioning. TLS client certificate supported in addition to or instead of password.

2.1 Last Will and Testament

The client registers a Last Will and Testament (LWT) message at connection time. If the station disconnects unexpectedly — due to power loss, WiFi drop, or firmware watchdog reset — the broker will publish the LWT payload to the state topic after the keep-alive timeout expires. Supervisory systems should subscribe to the state topic with a retained-message handler so that the last known state is immediately available on (re)subscribe.

LWT topic	<code>{base}/DATA/State</code>
LWT payload	<code>{"State": "ERROR"}</code>
LWT QoS	2 (exactly once)
LWT retain	true

3 Topic Hierarchy

All topics used by the LinFill-120 are relative to the {base} prefix. Each command has its own subscribe (CMD) and publish (DATA) topic pair; two additional DATA topics carry continuous state and streamed weight telemetry that are not tied to a single command.

Full Topic (relative to {base})	Direction	QoS	Retained?
/CMD/Start	Device subscribes	2	No
/DATA/Start	Device publishes	2	No
/CMD/Stop	Device subscribes	2	No
/DATA/Stop	Device publishes	2	No
/CMD/Home	Device subscribes	2	No
/DATA/Home	Device publishes	2	No
/CMD/Reset	Device subscribes	2	No
/DATA/Reset	Device publishes	2	No
/DATA/State	Device publishes	2	Yes
/DATA/CycleTime	Device publishes	2	No
/DATA/Weight	Device publishes	2	No

4 Command / Response Pattern

Every CMD payload is a JSON object containing at minimum a "Uuid" string (client-generated, unique per request). The station processes the command and publishes a JSON object to the matching DATA topic, echoing the same "Uuid" plus a "TimeStamp" (ISO 8601, UTC, millisecond precision, e.g. 2026-02-18T14:32:07.501Z) and a "State" field of either "SUCCESS" or "FAILURE". Commands received in an incompatible station state (e.g. Start while RUNNING) result in "FAILURE" with no state change. No parameters beyond "Uuid" are required for any of the four commands below.

Command	Valid when State is	Resulting state sequence	Notes
Start	IDLE	IDLE -> RUNNING -> IDLE	CycleTime and Weight streamed during the cycle; DATA/Start confirms cycle start, not completion
Stop	RUNNING	RUNNING -> IDLE	Halts cycle; lift retracts. CycleTime NOT published for the interrupted cycle.
Home	Any	* -> HOMING -> IDLE	Should be issued once on power-on before first Start
Reset	ERROR	ERROR -> IDLE	Clears fault flag; does not re-home

5 Payload Reference

5.1 Command Payloads ({base}/CMD/<Action>)

All four command topics share the same minimal payload shape:

Field	Type	Meaning
Uuid	string	Client-generated correlation identifier (UUIDv4 recommended). Echoed back in the DATA response.

Example — {base}/CMD/Start:

```
{"Uuid": "3f2a9e6c-1b4d-4e2a-9c7f-8a1d2e3f4b5c"}
```

5.2 Response Payloads ({base}/DATA/<Action>)

Field	Type	Meaning
Uuid	string	Copied unchanged from the triggering command payload
TimeStamp	string	ISO 8601 UTC timestamp with millisecond precision and 'Z' suffix
State	string	"SUCCESS" or "FAILURE"

Example — {base}/DATA/Start:

```
{"Uuid": "3f2a9e6c-1b4d-4e2a-9c7f-8a1d2e3f4b5c", "TimeStamp": "2026-02-18T14:32:07.501Z", "State": "SUCCESS"}
```

5.3 Telemetry Payloads

{base}/DATA/State carries the station's current operational mode. It is published on every state change (event-driven) and retained by the broker, so a new subscriber immediately receives the most recent value.

Payload value	Meaning	Entry conditions
"IDLE"	Ready; all actuators at rest	After successful Home; after cycle completion; after Stop; after Reset clears error
"RUNNING"	Fill cycle in progress	Start command accepted while State == IDLE
"ERROR"	Fault latched; motion inhibited	Limit-switch timeout; motor stall; L298N thermal shutdown; LWT published by broker
"HOMING"	Lift performing homing sequence	Home command accepted; also entered automatically at power-on

{base}/DATA/Weight streams the syringe weight reading continuously while State == RUNNING (not a single post-fill reading) so supervisory systems can chart the fill curve in real time. Payload: {"Weight": <grams, float>, "TimeStamp": <ISO 8601>, "Uuid": <the Start command's Uuid>}. From firmware v1.5.0, this requires an external balance connected to the ESP32 UART; the topic is silent if no balance is fitted.

{base}/DATA/CycleTime is published once per cycle, immediately after State transitions from RUNNING to IDLE following a successful fill. Payload: {"CycleTime_s": <seconds, float, 2 decimal places>, "TimeStamp": <ISO 8601>, "Uuid": <the Start command's Uuid>}. Not published if the cycle is interrupted by Stop or if the station faults during a cycle.

6 Security Recommendations

The default configuration (unencrypted TCP, username/password authentication) is adequate for isolated development networks. Before connecting the station to any production or GxP-regulated network, the following additional measures should be considered.

- **TLS encryption:** Enable MQTTS on port 8883. Provision the broker CA certificate onto the ESP32 via the provisioning UI. Consider mutual TLS (client certificate on the device) to authenticate the device to the broker.
- **Access control lists:** Restrict the client ID `linfill120-{serialNumber}` on the broker to publish/subscribe only to `{base}/CMD/#` and `{base}/DATA/#` and no other topics.
- **Network segmentation:** Place filling stations on a dedicated IoT VLAN with firewall rules restricting MQTT traffic to the authorised broker IP address only.
- **Credential rotation:** Change device passwords and rotate certificates at least annually, or following any personnel change that had access to the credentials.
- **Broker hardening:** Disable anonymous access on the broker. Log all connect and disconnect events. Set broker-side keep-alive expiry slightly longer than the device keep-alive (90 s is appropriate for a 60 s device keep-alive).
- **Uuid uniqueness:** Client Uuid values must be unique per outstanding request; reusing a Uuid before receiving its response will make correlation ambiguous.

7 Integration Checklist

The following checklist may be used as a quick commissioning aide-memoire. It does not replace the full OQ protocol.

#	Item	Done?
1	Broker reachable from device subnet; firewall permits TCP 1883 or 8883	■
2	Client ID <code>linfill120-{serialNumber}</code> is unique on the broker	■
3	<code>{base}</code> prefix agreed with namespace owner and recorded	■
4	LWT configured; test by powering off device and confirming <code>State=ERROR</code> appears on <code>DATA/State</code>	■
5	SCADA/MES subscribes to <code>DATA/State</code> with retained-message support	■
6	SCADA/MES subscribes to <code>DATA/CycleTime</code> and <code>DATA/Weight</code>	■
7	SCADA/MES generates a fresh Uuid per command and matches it against the <code>DATA/</code> response	■
8	All four commands tested; state transitions verified per Section 4	■
9	TLS and ACL configured for production use	■
10	Firmware version confirmed as 1.5.0 or later before go-live (required for <code>CMD/DATA</code> topic pattern)	■

8 Technical Support

For integration support, firmware files, or to report unexpected behaviour, contact Elara Automation GmbH using the details below. When raising a support ticket please include the unit serial number, the firmware version visible in the provisioning web interface, and a description of the MQTT broker software and version in use.

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